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SYSTEMATIC REVIEW

Epidemiology of Head Injuries Focusing on Concussions in Team Contact Sports: A Systematic Review

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Abstract

Background Although injuries to the head represent a small proportion of all sport injuries, they are of great concern due to their potential long-term consequences, which are even suspected in mild traumatic brain injuries.

Objective The aim of this review was to compare the incidence of concussions and other head injuries in elite level football, rugby, ice hockey and American Football.

Methods Four electronic databases (CINAHL, PsycINFO, Web of Science, PubMed) were searched. Prospective cohort studies on the incidence of concussion in elite athletes aged 17 years or older that were published in an English-language peer-reviewed journal since 2000 were included. Two authors independently evaluated study eligibility and quality. The extracted data on concussions were pooled in a meta-analysis using an inverse-variance fixed-effects model. The extracted data on head injuries were reported in a narrative and tabular summary.

Results The search yielded 7673 results of which 70 articles were included in the qualitative and 47 in the quantitative analysis. In our meta-analysis, we found the highest concussion incidences in rugby match play (3.89 and 3.00 concussions per 1000 h and athletic exposures (AEs), respectively), and the lowest in men's football training (0.01 and 0.08 per 1000 h and AEs, respectively). Overall, concussions and all head injuries were rare in training when compared to match play. Female players had an increased concussion risk in football and ice hockey when compared to male players.

Conclusion Future research should focus on concussion in women's contact sports, as there is little evidence available in this area. Methodological deficits are frequent in the current literature, especially regarding sample size and study power, and should be avoided.

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Key Points

When comparing football, rugby, ice hockey and American Football, the highest concussion incidence density was found in rugby, while the lowest was found in football.

Concussion incidence density varied across context (match vs. training) and between sexes, with female players at an increased concussion risk in ice hockey and football.

The most important methodological considerations regarding the current literature are insufficient sample size/study duration and heterogeneity in the reporting of outcomes.

1 Introduction

In recent years, sports-related concussions have been the topic of numerous studies, consensus statements, position papers and media reports [1–4]. The latest consensus statement defines concussions as a subset of mild traumatic brain injury (mTBI) without structural abnormalities on conventional neuroimaging modalities [1]. The most common acute symptoms of concussion are headache and dizziness, which usually resolve within 7–10 days [5]. However, in 10–15% of concussions persistent symptoms are reported [6]. Further, several pathological case-studies and case-series have suggested a potential for long-term health problems in athletes following a career in contact sports [7, 8]. These may include the development of neurodegenerative diseases (such as chronic traumatic encephalopathy [9], Alzheimer's dementia [10] or mild cognitive impairment [11]) and affective disorders (such as depression [12]). However, an association between high exposure to head impacts and late-life sequelae remains controversial since causality has not been established [13].

Through extensive media coverage of the aforementioned findings [14], concussions have become a matter of great public concern with practical implications for sports participation [15, 16]. However, the magnitude of the problem and factors associated with concussion risk are not yet well understood. Several studies suggest that athletes are at a higher risk of incurring concussions when participating in match play [17] or if they have a history of previous concussions [18]. Some authors also found the female sex to be associated with an increased risk of concussion [19, 20], where others did not [21, 22]. Contact sports, such as ice hockey, American Football and rugby are thought to have a particularly high concussion risk [2]. Further, in football (soccer), heading the ball is considered to increase the risk for concussions, due to incidental head-to-head or elbow-to-head contact [23]. Koh et al. systematically reviewed the incidence of concussion in contact sports in 2003 [24]. Overall, the authors found the highest concussion incidence in men's ice hockey, followed by American Football, rugby and football (soccer). This was surprising, considering that protective headgear is mandatory only in the two sports reporting the highest incidence rates.

However, in the past decade numerous studies on sports concussion have been published and to the authors' knowledge no recent systematic review and meta-analysis exists that quantifies and compares concussion incidences in contact sports. Therefore, the primary aim of this systematic review is to summarise recent evidence on the incidence of concussion in football (soccer), rugby (union), ice hockey and American Football. A secondary aim is to compare the incidence of all head injuries (including concussions) in these four sports, since concussions are suspected to be under-diagnosed [25, 26]. Moreover, this review will analyse factors that may modulate injury incidence, such as sex, sport and context (match vs. training). Since the field of concussion epidemiology faces several challenges, another aim is to highlight and give recommendations on key methodological issues in this area of research. These include, but are not limited to, different concussion definitions and the lack of a gold standard diagnostic procedure [27, 28].

2 Methods

2.1 Search Strategy

In order to identify relevant studies an online search of four databases (PubMed, CINAHL, Web of Science, PsycINFO) was conducted. Papers published between 2000 and 29th June 2016 were identified using the following search strategy: (soccer OR football OR American Football OR Ice Hockey OR Rugby) AND (concussion OR brain injur* OR head injur* OR injur* OR athletic injur* OR sports concussion OR sports related concussion OR mild traumatic brain injury OR mTBI OR traumatic brain injury OR TBI OR craniocerebral trauma OR brain damage) AND (prevalence OR epidemiolog* OR descriptive OR overview OR incidence OR classification OR numerical data). The search was conducted by two authors (AP and AG) independently. In addition, hand searching of key primary and review articles was carried out.

2.2 Eligibility Criteria

Studies that met the following criteria were included in the review:

- Types of participants: study conducted within high-level (at least national level) football, rugby, ice hockey or American Football players aged 17 years or older.
- Types of data: reporting incidence density of concussion, or data on total sports exposure and number of concussions.
- Types of studies: full-text prospective cohort study published in English in a peer-reviewed journal since 2000.

Studies that met the following criteria were excluded from the review:

- Types of participants: participant characteristics regarding age, sex, type of sport or level of play were not described.
- Types of data: the presented data were used in another article included in the analysis (i.e. duplicate publication).

2.3 Quality Rating

All studies were evaluated with respect to potential for bias in assessing concussion incidence. Thus, the quality rating reflects the study quality regarding the accuracy of the published data on concussion incidence only. To this end we adapted a rating scale by van der Windt et al. [29], developed for their review of studies focusing on injury surveillance data. The adapted scale contains 10 items, and a maximum rating score of 20 points was achievable (Electronic Supplementary Material Appendix S1). The assessment was carried out by two reviewers (AP and AG) independently and, in case of disagreement, consensus was achieved through discussion with the fourth author (AJ). When it was unclear whether a criterion was met, no points were awarded. A summary score was calculated and expressed as the percentage of the total points scored. All studies were then split into three groups based on their quality rating ($\geq 70\%$ high, $\geq 50\%$ moderate, $< 50\%$ low), in order to analyse methodological differences. Low-quality studies were excluded from the narrative and tabular summary, but not from the meta-analysis.

2.4 Data Extraction

Data were collected on the study design, participant characteristics (sex, age, sport, level of play), sample size, injury definition and method of injury assessment as well as season, training and competition injury incidence. All data were extracted by two researchers (AP and AG) independently using a standardised data extraction tool.

2.5 Data Synthesis

Data of moderate and high-quality studies are presented separately for all head injuries and concussions only in a narrative and tabular summary. Incidence rates are grouped by sport, sex, context, exposure and injury definition (time-loss vs. medical attention). Where standardised injury or exposure information was missing, incidences and confidence intervals were calculated from raw data. Rate ratios were calculated to quantify risk increases in relation to sex (female/male) and context (match/training). All studies that reported concussion data for match and/or training exposure were further included in a meta-analysis. Season incidence rates were not pooled, as they are highly dependent on the ratio of match-to-training exposure. Heterogeneity in the meta-analysis was assessed using I^2 statistics, which indicate the percentage of variation attributed to heterogeneity. Even though heterogeneity was high, a fixed-effects model was deemed appropriate, as small studies tend to over- or underestimate the incidence of relatively rare events, such as concussions. Pooled incidence rates were calculated for all four sports and grouped by sex and exposure definition. Statistical analysis was carried out using an open source spreadsheet in Microsoft Excel 2016 [30] and in IBM SPSS Statistics 24. All data are reported according to the 'Meta-analysis Of Observational Studies in Epidemiology' (MOOSE) guideline [31].

3 Results

Our initial literature search yielded 7673 results. After title and abstract screening, 283 full-text articles were reviewed. 203 of these articles did not meet the inclusion criteria and ten met at least one exclusion criterion. The most frequent reason for exclusion at this stage was inadequate or no data on exposure and/or concussions. Finally, 70 articles were included, of which 47 were further summarised in a meta-analysis (Fig. 1).

3.1 Study Characteristics

The included studies reported on a total sports exposure of 92,715 h and 35,899,538 athletic exposures (AEs). Twenty-eight studies were conducted in football [32–59], 14 in ice hockey [60–73], 14 in American Football [74–87] and 12 in rugby [21, 88–98]. Two studies contained data on multiple sports [99, 100]. Most studies ($n = 31$) focused on collegiate level athletes, a quarter of studies ($n = 17$)

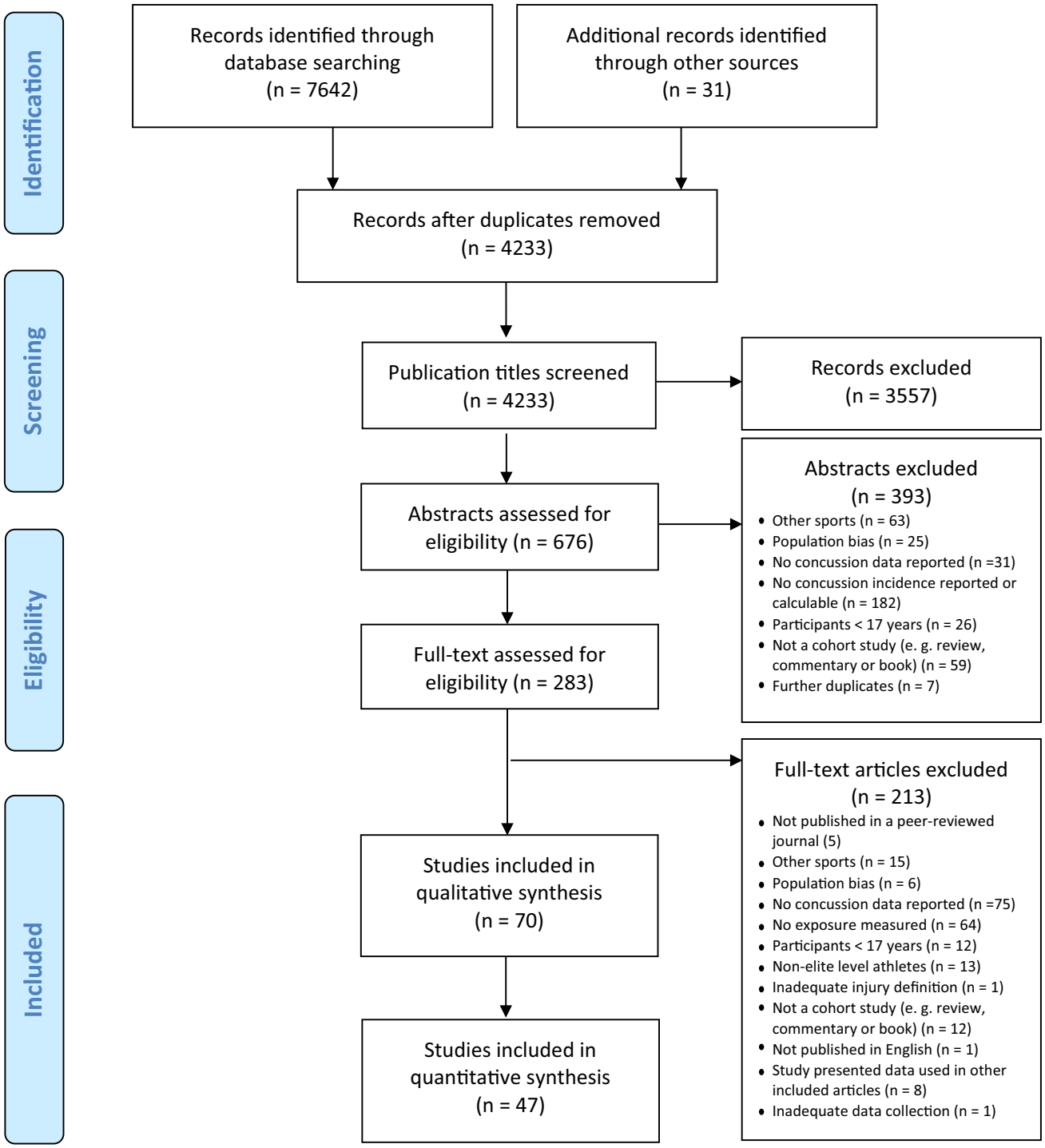

Fig. 1 PRISMA flow diagram

investigated international level athletes and a third of studies ($n = 22$) were conducted with professional athletes. Data on female athletes were presented in 30% of studies ($n = 21$), while data on male athletes were presented in 86% of studies ($n = 60$). Further, notably more studies assessed match exposure ($n = 47$) than training exposure ($n = 27$). The most commonly employed exposure definition was athlete hours (h, $n = 37$) followed by athlete exposures (AEs, $n = 28$); one study reported injury rates for both measures [21]; four studies used team/player games as exposure measure [55, 68, 72, 73]. Injury definition varied in specificity among studies; however, the majority used a time-loss based definition (TL, $n = 40$), while most other studies ($n = 29$) defined an

injury as any sport-related event or any event that required medical attention (MA). In our review, we treat the latter two definitions as equal, since medical personnel collected most of the data, which implies medical attention when an injury is reported [101, 102]. One study failed to provide an injury definition [78] and a concussion-specific injury definition was not reported in most studies.

3.2 Study Quality

The average study quality score was 57% (SD = 13.3%) ranging from 30 to 80% (Table 1). There were three predominant methodological issues: (1) over three-quarters of the studies did not carry out a baseline assessment regarding concussion history; (2) two-thirds of studies did not sufficiently describe the study population regarding sample size and age; (3) data collection intervals in about half of studies were ≥ 1 month long or not reported at all. When grouped by quality rating the main differences between high ($n = 17$; mean: 74%) and poor ($n = 17$; mean: 39%) quality studies were the total exposure time measured (partly due to item weighing) and whether the data collection personnel were briefed and medically trained.

3.3 Concussions

Of the 70 included studies, 53 were evaluated as moderate or high quality (Table 2) and thus included in the qualitative data synthesis on concussion incidence density (CID). Forty-seven studies reported data for match or training exposure and were pooled in the meta-analysis (Table 3).

3.3.1 Qualitative Synthesis

CID in football ranged from 0.004 [56] to 2.44 [52] per 1000 h. The median season, match and training incidence rates per 1000 h were, respectively, 0.13 [38], 1.25 [42, 45] and 0.08 [38]. Only three studies reported CID in football players per 1000 AEs with rates ranging from 0.08 [32] to 1.42 [36]. The median season, match and training incidence rates per 1000 AEs were, respectively, 0.38 [36, 100], 1.03 [61, 100] and 0.15 [36, 100]. In rugby CID ranged from 0.02 [95] to 8.93 [90] per 1000 h and from 0.29 [21] to 9.64 [99] per 1000 AEs. The median season and match incidence rates per 1000 h/AEs were 0.53/0.90 [21, 91, 98] and 4.33/2.87 [21, 93, 95], respectively. Very little information on concussions in rugby training was available; the two studies evaluating training injury incidence per 1000 h had an average rate of 0.07 [91, 95] and the median CID per 1000 AEs was 0.33 [99]. In ice hockey CID per 1 000 h/AEs ranged from 0.09/0.10 [61, 69] to 5.13/14.93 [65, 72]; however, studies reporting rates per 1000 h were rare. The median season, match and training incidence rates per 1000 AEs were 0.80 [62, 100], 2.62 [60, 100] and 0.28 [100], respectively. In American Football, there were no studies reporting incidence rates per 1000 h; CID per 1000 AEs ranged from 0.18 [77] to 5.56 [74], with median season, match and training incidence rates of 0.71 [74, 100], 3.60 [76, 85] and 0.41 [83, 100], respectively.

Overall, training concussions were rare when compared to match concussions in all sports; the median rate ratios between match and training in football, rugby, ice hockey and American Football were 12.7 [32, 36], 21.7 [91], 9.15 [60, 100] and 8.11 [79], respectively. When comparing male and female players, relevant differences were found in football and rugby. While in football female players were more likely to incur a concussion in both match and training (rate ratios 1.00–3.33) [32, 36, 40, 42, 44, 45, 100], a study in rugby [21] found the opposite (rate ratios 0.74–0.98). A less consistent picture was found in ice hockey, where two authors found male [72, 73, 100] and one study found female players [60] to have a higher match CID. Training ice hockey concussion incidence in both male and female players was only assessed in one study [100], which reported female players to have a slightly higher injury risk with a rate ratio of 1.2.

3.3.2 Quantitative Synthesis

Of the 47 studies included in the meta-analysis, 21 studies reported data on matches only [33–35, 39, 45, 51–55, 59, 63, 65, 68, 72, 78, 85, 88, 90, 93, 96], two on training only [44, 83] and 24 assessed both [21, 32, 36, 38, 40, 42, 56, 57, 60, 61, 66, 69, 74–77, 79, 82, 91, 92, 94, 95, 99, 100]. Further, 37 studies provided information on male players only [32–35, 38–40, 52–54, 56, 57, 59, 61, 63, 66, 68, 69, 72, 74–79, 82, 83, 85, 88, 90–96, 99], four on female players only [36, 51, 55, 60] and six studies assessed both [21, 42, 44, 45, 65, 100]. There were insufficient data on female rugby players to be included in the meta-analysis. In a match context, the lowest pooled incidence rates were found in men's football (0.44/1000 h, 1.07/1000 AEs) and the highest in men's rugby (3.89/1000 h, 3.00/1000 AEs) regardless of exposure definition. Similarly, in a training context the lowest pooled incidence rates were found in men's football (0.01/1000 h, 0.08/1000 AEs) regardless of exposure definition and the highest CID per 1000 AEs was found in men's rugby (CID = 0.37). However, the highest incidence rate per 1000 h of training was found in women's football (CID = 0.13). When comparing pooled incidence rates by sex in football and ice hockey, female players had consistently higher CIDs regardless of context and exposure definition. Rate ratios in football ranged from 1.38 in matches to 13.00 in training; rate ratios in ice hockey were 1.39 in matches and 2.58 in training.

Table 1 Study characteristics

Study	Type of sport, level of play, context	Setting, timeframe, duration	Athletes (*n*), sex, average age (years)	Exposure definition, injury definition, injury type	Data collected by, follow-up	Study quality (%)
Football						
Agel et al. [32]	F, Col, S M T	NCAA DIV I–III, 1988–2002, 15 seasons	N/A, m, N/A	AE, TL, C and HI	Athletic trainer, weekly	65
Akodu et al. [33]	F, Int, M	WAFU Cup, 2011, 1 tournament	82, m, 20.8	h, MA, C and HI	Physiotherapist, end of tournament	40
Andersen et al. [34]	F, Pro, M	Norwegian and Icelandic 1st League, 1999–2000, 2 seasons	N/A, m, N/A	h, TL, C and HI	Medical staff, monthly	30
Bjorneboe et al. [35]	F, Pro, M	Norwegian 1st League, 2010–2011, 2 seasons	N/A, m, N/A	h, TL, C and HI	Medical staff, monthly	45
Dick et al. [36]	F, Col, S M T	NCAA DIV I–III, 1988–2002, 15 seasons	N/A, f, N/A	AE, TL, C and HI	Athletic trainer, weekly	65
Dupont et al. [37]	F, Pro, S	Scottish Premier League team, 2007–2008, 2 seasons	32, m, 25.6	h, TL, C and HI	Club physician, N/A	45
Dvorak et al. [39]	F, Int, M	World Cup, 2006, 1 tournament	N/A, m, N/A	h, MA, C and HI	Team physician, every game	50
Dvorak et al. [38]	F, Int, S M T	World Cup, 2010, 1 tournament	736, m, N/A	h, MA, C and HI	Team physician, every game	65
Eirale et al. [40]	F, Pro, S M T	Qatar 1st League, 2008, 1 season	230, m, 28.4	h, TL, C and HI	Head of medical staff, daily	75
Eirale et al. [41]	F, Pro, S	Qatar 1st League, 2008–2011, 3 seasons	527, m, 25.8	h, TL, C and HI	Physician or physiotherapist, daily	70
Ekstrand et al. [42]	F, Pro, S M T	Elite level UEFA teams, 2003–2008, 6 seasons	767, m (613) and f (154), 25 (m) and 23 (f)	h, TL, C and HI	Team medical staff, monthly	75
Ekstrand et al. [43]	F, Pro, S	Elite level UEFA teams, 2003–2009, 7 seasons	2226, m, 25.7	h, TL, C	Team medical staff, monthly	70
Fuller et al. [44]	F, Col, S T	NCAA, 2005–2006, 2 seasons	N/A, m and f, N/A	h, TL, C and HI	Athletic trainer, weekly	60
Fuller et al. [45]	F, Col, S M	NCAA, 2005–2006, 2 seasons	N/A, m and f, N/A	h, TL, C and HI	Athletic trainer, weekly	60
Gaulrapp et al. [46]	F, Pro, S	German Women's Premier League, N/A, 1 season	254, f, 22.8	h, TL, C and HI	Team physician, monthly	60
Giza et al. [47]	F, Pro, S	WUSA, 2001–2002, 2 seasons	202, f, N/A	h, MA, C and HI	Team trainer and verified by team physician, N/A	45
Hagglund et al. [48]	F, Pro, S	Swedish Premier League, 2005, 1 season	467, m (239) and f (228), 25 (m) and 23 (f)	h, TL, C and HI	Team medical staff, monthly	75
Hassabi et al. [49]	F, Pro, S	Iranian Premier League club, 2005–2006, 4 months	21, m, 24	h, TL, C and HI	Team physicians, daily	35
Jacobson et al. [50]	F, Pro, S M T	Swedish female Premier League, 2000, 1 season	269, f, 23	h, TL, C and HI	Trainer/coach, weekly	60
Junge and Dvorak [51]	F, Int, M	International tournaments, 1999–2006, 8 tournaments	N/A, f, N/A	h, MA, C and HI	Team physician, every game	70

Table 1 continued

Study	Type of sport, level of play, context	Setting, timeframe, duration	Athletes (*n*), sex, average age (years)	Exposure definition, injury definition, injury type	Data collected by, follow-up	Study quality (%)
Junge and Dvorak [52]	F, Int, M	World Cup, 2014, 1 tournament	N/A, m, N/A	h, MA, C and HI	Team physician, every game	50
Junge et al. [53]	F, Int, M	World Cup, 2002, 1 tournament	N/A, m, N/A	h, MA, C and HI	Team physician, every game	55
Junge et al. [54]	F, Int, M	International tournaments, 1998–2001, 8 tournaments	N/A, m and f, N/A	h, MA, C and HI	Team physician, every game	50
Meyers [55]	F, Col, M	NCAA DIV I, 2007–2011, 5 seasons	N/A, f, N/A	TG, MA, C	Head athletic trainer and team physician, < a week after match	55
Nilsson et al. [56]	F, Pro, S M T	UEFA, 2001–2009, 9 seasons	1401, m, 25.8	h, TL, C and HI	Club medical officer, monthly	65
Owen et al. [57]	F, Pro, S M T	Elite soccer players, N/A, 2 seasons	23, m, 26.8	h, TL, C and HI	Medical staff, N/A	35
Walden et al. [58]	F, Int, S	EURO and WOCO, 2004–2005, 2 tournaments	528, m (368) and f (160), N/A	h, TL, C	Team physician, every 3 days	45
Yoon et al. [59]	F, Int, M	Asian Cup and U19 Asian Championship, 2000, 2 tournaments	411, m, 22	h, MA, C and HI	Team physicians, every game	50
Ice hockey						
Agel and Harvey [62]	IC, Col, S	NCAA DIV I and III, 2000–2006, 7 seasons	N/A, m and f, N/A	AE, TL, C	Athletic trainer, weekly	65
Agel et al. [60]	IC, Col, S M T	NCAA DIV I–III, 2000–2003, 4 seasons	N/A, f, N/A	AE, TL, C and HI	Athletic trainer, weekly	60
Agel et al. [61]	IC, Col, S M T	NCAA DIV I–III, 1988–2003, 16 seasons	N/A, m, N/A	AE, TL, C and HI	Athletic trainer, weekly	60
Benson et al. [64]	IC, Col, S	CIAU, 1997–1998, 1 season	642, m, 22	AE, MA, C	Team physician and/or doctor, N/A	75
Benson et al. [63]	IC, Pro, M	NHL, 1997–2003, 7 seasons	N/A, m, N/A	h, MA, C	Team physicians, N/A	60
Echlin et al. [65]	IC, Col, M	Canadian Varsity Ice-hockey, 2011–2012, 1 season	45, m (25) and f (20), N/A	AE, MA, C	Independent observers (1 physician and 1 non-physician), every game	50
Flik et al. [66]	IC, Col, S M T	Eastern College DIV I, 2001–2002, 1 season	12 teams, m, N/A	AE, TL, C	Athletic trainer, monthly	45
Grant et al. [67]	IC, Col, S	Varsity Team DIV I, 2003–2010, 8 seasons	79, m, 20.2	AE, TL, C and HI	Athletic trainer, N/A	40
Hutchison et al. [68]	IC, Pro, M	NHL, 2006–2009, 3.5 seasons	N/A, m, N/A	TG, MA, C	Athletic trainer or therapist and diagnosis by physician, N/A	45
Kuzuhara et al. [69]	IC, Elite, S M T	Japanese ice-hockey team, 2002–2005, 3 seasons	94, m, 25–27	h, MA, C and HI	Athletic trainer, N/A	55
Rishiraj et al. [70]	IC, Col, S	CWUAA, 1991–1996, 6 seasons	150, m, 22	AE, TL, C and HI	Team athletic therapist, weekly	55

Table 1 continued

Study	Type of sport, level of play, context	Setting, timeframe, duration	Athletes (*n*), sex, average age (years)	Exposure definition, injury definition, injury type	Data collected by, follow-up	Study quality (%)
Schick and Meeuwisse [71]	IC, Col, S	CWUAA, 1998–1999, 1 season	261, m (147) and f (114), 23.5 (m) and 20.5 (f)	AE, TL, C and HI	Team athletic therapist, fortnightly	50
Tuominen et al. [72]	IC, Int, M	World Cup and OWG, 2007–2013, 41 tournaments	6666, m, N/A	PG, MA, C and HI	Team physician, every game	65
Tuominen et al. [73]	IC, Int, M	Women's World Cup and OWG, 2007–2014, 25 tournaments	3376, f, N/A	PG, MA, C and HI	Team physician, every game	65
American Football						
Booher et al. [74]	AF, Col, S M T	NCAA DIV I, 2001–2002, 1 season	N/A, m, N/A	AE, MA, C	Athletic trainer, weekly	75
Dick et al. [75]	AF, Col, S M T	NCAA DIV I–III, 1988–2004, 16 seasons	N/A, m, N/A	AE, TL, C	Athletic trainer, weekly	65
Dompier et al. [76]	AF, Col, S M T	NCAA DIV I–III, 2012–2013, 2 seasons	N/A, m, N/A	AE, MA, C	Athletic trainer, daily	80
Feeley et al. [77]	AF, Pro, S M T	NFL preseason training camp, 1988–2007, 5 weeks (including 4-5 games)	696, m, 25.8	AE, TL, C	Author, daily	70
Funk et al. [78]	AF, Col, M	Virginia Tech College Football, 2006–2010, 5 seasons	98, m, N/A	AE, N/A, C	N/A	30
Guskiewicz et al. [80]	AF, Col, S	NCAA DIV I–III, 1995–1997, 3 seasons	11,460, m, N/A	AE, MA, C	Athletic trainer, at completion of season	65
Guskiewicz et al. [79]	AF, Col, S M T	NCAA DIV I–III, 1999–2001, 3 seasons	2905, m, N/A	AE, MA, C	Athletic trainer, N/A	65
Hootman et al. [81]	AF, Col, S	NCAA, 1988–2003, 16 seasons	N/A, m, N/A	AE, TL, C	Athletic trainer, weekly	65
Iguchi et al. [82]	AF, Col, S M T	Japanese DIV I College Football team, 2007–2009, 3 seasons	289, m, 20.5	AE, TL, C and HI	Athletic trainer, N/A	45
Kerr et al. [83]	AF, Col, T	NCAA, 2004–2008, 5 seasons	N/A, m, N/A	AE, MA, C	Athletic trainer, in real time through a web-based platform	80
Kilcoyne et al. [84]	AF, Col, S	NCAA DIV I, 2009–2010, 2 seasons	3 teams, m, 18–22	AE, TL, C	Team athletic trainer, N/A	50
Lynall et al. [85]	AF, Col, M	NCAA DIV I, 2009–2013, 5 seasons	N/A, m, N/A	AE, MA, C	Athletic trainer, in real time through an electronic health record application	75
Meeuwise et al. [86]	AF, Col, S	CWUAA, 1993–1997, 5 seasons	881, m, N/A	AE, MA, C and HI	Athletic therapist, fortnightly	75
Zemper [87]	AF, Col, S	NCAA, 1997–1998, 2 seasons	N/A, m, N/A	AE, TL, C	Team athletic trainer, weekly	75
Rugby						

Table 1 continued

Study	Type of sport, level of play, context	Setting, timeframe, duration	Athletes (*n*), sex, average age (years)	Exposure definition, injury definition, injury type	Data collected by, follow-up	Study quality (%)
Best et al. [88]	R, Int, M	IRB World Cup, 2003, 1 tournament	N/A, m, N/A	h, TL, C and HI	Team medical officers, N/A	30
Brooks et al. [89]	R, Int, S	English Rugby Squad, 2002–2003, 63 weeks	63, m, N/A	h, TL, C and HI	Team physician, N/A	35
Cross et al. [90]	R, Pro, M	English Premiership, 2012–2013, 2 seasons	810, m, N/A	h, MA, C	Team medical staff, N/A	80
Fuller et al. [91]	R, Int, SMT	IRB World Cup, 2007, 1 tournament	626, m, 27.6	h, TL, C and HI	Team physician, N/A	50
Fuller et al. [92]	R, Int, S M T	IRB World Cup, 2011, 1 tournament	615, m, 27.4	h, TL, C and HI	Team physician or physiotherapist, N/A	55
Fuller et al. [93]	R, Pro, M	English Premiership, 2007–2013, 3 seasons	N/A, m, N/A	h, TL, C	Team physician, N/A	65
Fuller et al. [93]	R, Int, M	Pacific Nations Cup, 2007–2013, 2 tournaments	N/A, m, N/A	h, TL, C	Team physician, N/A	45
Holtzhausen et al. [94]	R, Pro, S M T	Super 12 competition, 1999, 1 tournament	75, m, N/A	h, TL, C and HI	Team physician, weekly (phone)	40
Kemp et al. [95]	R, Pro, S M T	English Premiership, 2002–2003 and 2005, 3 seasons	757, m, N/A	h, TL, C and HI	Medical staff, weekly	70
Kerr et al. [21]	R, Col, S M T	English NERFU, 2005, 1 season	66 teams, m and f, 17–21	h and AE, MA, C and HI	Medical staff, coach or identified player, weekly	70
Moore et al. [96]	R, Int, M	Welsh National Team, 2012–2014, 3 seasons	78, m, N/A	h, TL, C and HI	Team physical therapist, N/A	45
Peck et al. [97]	R, Col, S	Military Academy College, 2006–2010, 5 seasons	369, m (240) and f (129), N/A	AE, MA, C and HI	Athletic trainer, N/A	55
Schick et al. [98]	R, Int, S	IRB Women's World Cup, 2006, 1 tournament	339, f, 28	h, MA, C and HI	Team physician, collected at regular intervals	50
Multi-sport						
Willigenburg [99]	R and AF, Col, SMT	NCAA DIV I, 2012–2014, 3 seasons	2 teams, m, N/A	AE, TL, C and HI	Medical professionals, daily	55
Zuckerman et al. [100]	F, IC and AF, Col, S M T	NCAA, 2009–2013, 5 seasons	N/A, m, N/A	AE, MA, C	Athletic trainer, real time through a health record application	55

F football, *R* rugby, *IC* ice hockey, *AF* American football, *Col* collegiate, *Pro* professional, *Int* international, *S* season, *M* match, *T* training, *NCAA* National Collegiate Athletic Association, *DIV* division, *UEFA* Union of European Football Associations, *WUSA* Women's United Soccer Association, *NERFU* New England Rugby Football Union, *EURO* Men's European Championship, *WOCO* Women's European Championship, *WAFU* West African Football Union, *IRB* International Rugby Board, *NHL* National Hockey League, *NFL* National Football League, *CIAU* Canadian Intercollegiate Athletic Union, *CWUAA* Canada West Universities Athletic Association, *OWG* Olympic World Games, *N/A* not available, *m* male, *f* female, *h* hours, *AE* athletic exposures, *PG* per game, *TG* team game, *TL* time loss, *MA* medical attention, *C* concussion, *HI* head injury

Table 2 Concussion incidence per 1000 exposures in moderate and high-quality studies

Sex, sport	Exposure definition	Season CID (95% CI)	Match CID (95% CI)	Training CID (95% CI)
Men's football	Hours	0.13 (0.00–0.31) [38][a]	0.49 (0.00–1.45) [38][a]	0.08 (0.00–0.24) [38][a]
		0.05 (0.00–0.12) [40]	0.47 (0.00–1.39) [39][a]	0.03 (0.00–0.09) [40]
		0.03 (0.00–0.06) [41]	0.16 (0.00–0.47) [40]	0.04 (0.01–0.07) [42]
		0.13 (0.08–0.18) [42]	0.62 (0.34–0.90) [42]	0.10 (0.06–0.14) [44]
		0.06 (0.04–0.08) [43]	1.57 (1.16–1.99) [45]	0.004 (0.00-0.01) [56]
		0.27 (0.21–0.33) [44, 45]	2.44 (0.30–4.58) [52][a]	
		0.10 (0.03–0.17) [48]	1.89 (0.04–3.74) [53][a]	
		0.06 (0.04–0.08) [56]	0.76 (0.09–1.43) [54][a]	
			0.35 (0.25–0.45) [56]	
			1.82 (0.00–3.88) [59][a]	
	AEs	0.28 (0.25–0.30) [32]	1.08 (0.97–1.18) [32]	0.08 (0.06–0.09) [32]
		0.34 (0.25–0.44) [100][a]	0.97 (0.64–1.30) [100][a]	0.18 (0.10–0.25) [100][a]
Women's football	Hours	0.23 (0.09–0.37) [42]	0.93 (0.24–1.62) [42]	0.10 (0.00–0.20) [42]
		0.42 (0.35–0.49) [44, 45]	2.15 (1.71–2.58) [45]	0.14 (0.10–0.19) [44]
		0.17 (0.08–0.27)[46]	2.09 (0.91–3.27) [51][a]	
		0.11 (0.02–0.20) [48]	1.63 (1.15–2.12) [55][a]	
		0.14 (0.03–0.17) [50]		
	AEs	0.41 (0.38–0.44) [36]	1.42 (1.29–1.55) [36]	0.12 (0.10–0.14) [36]
		0.63 (0.52–0.74) [100][a]	0.97 (0.64–1.30) [100][a]	0.21 (0.14–0.28) [100][a]
Men's rugby	Hours	0.37 (0.10–0.64) [91]	2.16 (1.49–2.83) [21][a]	0.12 (0.00–0.28) [91]
		0.85 (0.42–1.29) [92]	8.93 (7.63–10.23) [90][a]	0.02 (0.00–0.03) [95]
		0.31 (0.25–0.37) [95]	2.60 (0.32–4.89) [91]	
			7.81 (3.86–11.77) [92]	
			4.60 (3.85–5.35) [93]	
			5.36 (0.00–11.42) [93]	
			4.06 (3.25–4.87) [95]	
	AEs	1.28 (0.92–1.64) [21][a]	2.87 (1.98–3.76) [21][a]	0.37 (0.13–0.61) [21][a]
		0.33 (0.23–0.43) [97][a]	9.64 (3.34–15.94) [99]	0.33 (0.00–0.96) [99]
		2.50 (0.95–4.04) [99]		
Women's rugby	Hours	0.68 (0.01–1.35) [98][a]	1.58 (0.96–2.21) [21][a]	
	AEs	0.90 (0.59–1.21) [21][a]	2.11 (1.28–2.93) [21][a]	0.29 (0.08–0.51) [21][a]
		0.44 (0.28–0.59) [97][a]		
Men's ice hockey	Hours	0.36 (0.04–0.67) [69][a]	1.80 (1.65–1.95) [63]	0.09 (0.0–0.3) [69][a]
			1.56 (0.03–3.08) [69][a]	
			5.13 (3.74–6.53) [72][a]	
	AEs	0.41 (0.37–0.44) [61]	1.49 (1.34–1.63) [61]	0.10 (0.08–0.12) [61]
		0.70 (0.62–0.78) [62]	7.50 (0.15–14.86) [65][a]	0.25 (0.18–0.32) [100][a]
		1.55 (1.21–1.89) [64]	2.49 (2.11–2.86) [100][a]	
		0.50 (0.29–0.70) [70]		
		0.97 (0.51–1.43) [71]		
		0.79 (0.69–0.89) [100][a]		
Women's ice hockey	Hours		2.33 (0.95–3.71) [73][a]	
	AEs	0.91 (0.71–1.11) [60]	2.75 (2.04–3.47) [60]	0.33 (0.19–0.47) [60]
		0.81 (0.73–0.91) [62]	14.93 (2.98–26.87) [65][a]	0.30 (0.18–0.42) [100][a]
		1.18 (0.45–1.93) [71]	2.01 (1.50–2.52) [100][a]	
		0.75 (0.59–0.91) [100][a]		

Table 2 continued

Sex, sport	Exposure definition	Season CID (95% CI)	Match CID (95% CI)	Training CID (95% CI)
Men's American Football	AEs	0.74 (0.66–0.81) [74][a]	5.56 (4.88–6.24) [74][a]	0.25 (0.20–0.35) [74][a]
		0.43 (0.42–0.45) [75]	2.34 (2.24–2.45) [75]	0.27 (0.26–0.28) [75]
		0.83 (0.73–0.93) [76][a]	3.74 (3.04–4.43) [76][a]	0.53 (0.45–0.61) [76][a]
		0.45 (0.25–0.66) [77]	3.25 (1.41–5.09) [77]	0.18 (0.04–0.30) [77]
		0.53 (0.48–0.57) [79][a]	3.81 (3.04–4.57) [79][a]	0.47 (0.38–0.56) [79][a]
		0.81 (0.70–0.93) [80][a]	3.45 (2.93–3.97) [85][a]	0.39 (0.36–0.42) [83]
		0.37 (0.36–0.38) [81]	4.35 (1.34–7.36) [99]	0.82 (0.48–1.16) [99]
		0.85 (0.64–1.05) [84]	3.01 (2.64–3.37) [100][a]	0.42 (0.38–0.46) [100][a]
		1.01 (0.81–1.21) [86][a]		
		0.57 (0.51–0.63) [87]		
		1.04 (0.67–1.41) [99]		
		0.67 (0.62–0.72) [100][a]		

CID concussion incidence density, *CI* confidence interval, *AEs* athletic exposures

[a]Medical attention injury definition

Table 3 Pooled match and training concussion incidence per 1000 exposures

Sex, sport	Exposure definition	Total exposure	Total concussions	*df*	CID (95% CI)	SE	I^2 (%)
Match							
Men's football	Hours	241,746	153	13	0.44 (0.36–0.53)	0.04	71
	AEs	392,536	420	1	1.07 (0.97–1.17)	0.05	0
Women's football	Hours	83,838	157	3	1.76 (1.48–2.05)	0.14	68
	AEs	377,732	564	1	1.48 (1.35–1.6)	0.06	84
Men's rugby	Hours	102,081	502	9	3.89 (3.51–4.28)	0.20	91
	AEs	14,877	49	1	3.00 (2.12–3.88)	0.45	77
Men's ice hockey	Hours	374,824	875	3	2.01 (1.87–2.15)	0.07	98
	AEs	358,128	613	3	1.63 (1.49–1.76)	0.07	90
Women's ice hockey	AEs	50,958	123	2	2.27 (1.86–2.69)	0.21	72
Men's American Football	AEs	1,136,041	3005	9	2.52 (2.43–2.62)	0.05	94
Training							
Men's football	Hours	1,147,939	38	4	0.01 (0–0.01)	0.00	87
	AEs	1,563,973	135	1	0.08 (0.07–0.1)	0.01	84
Women's football	Hours	321,360	44	1	0.13 (0.09–0.17)	0.02	0
	AEs	1,276,398	165	1	0.12 (0.1–0.14)	0.01	85
Men's rugby	Hours	317,847	7	1	0.02 (0–0.03)	0.01	32
	AEs	27,351	10	1	0.37 (0.14–0.59)	0.12	0
Men's ice hockey	AEs	1,242,178	163	2	0.12 (0.1–0.14)	0.01	89
Women's ice hockey	AEs	149,129	47	1	0.31 (0.22–0.4)	0.05	0
Men's American Football	AEs	14,607,661	4482	8	0.30 (0.29–0.31)	0.00	95

AEs athletic exposures, *df* degrees of freedom, *CID* concussion incidence density, *CI* confidence interval, *SE* standard error, I^2 measure of inconsistency (i.e. heterogeneity)

3.4 All Head Injuries

Of the 70 included studies, 46 provided data on incidence density of all head injuries including concussions (HIID) and 33 of these studies were evaluated as moderate or high quality (Table 4). HIID in football ranged from 0.02 [56] to 13.74 [54] per 1000 h. The median season, match and training incidence rates per 1000 h were 0.25 [41], 3.28

Table 4 Head injury incidence per 1000 exposures in moderate- and high-quality studies

Sex, sport	Exposure definition	Season HIID (95% CI)	Match HIID (95% CI)	Training HIID (95% CI)
Men's football	Hours	1.12 (0.59–1.65) [38][a]	6.35 (2.90–9.80) [38][a]	0.30 (0.01–0.59) [38][a]
		0.10 (0.00–0.20) [40]	6.15 (2.86–9.49) [39][a]	0.03 (0.00–0.09) [40]
		0.25 (0.17–0.33) [41]	0.48 (0.00–1.02) [40]	0.18 (0.12–0.24) [42]
		0.30 (0.23–0.33) [42]	0.97 (0.62–1.32) [42]	0.21 (0.15–0.26) [44]
		0.51 (0.43–0.59) [44, 45]	3.28 (2.75–3.81) [45]	0.02 (0.01–0.03) [56]
		0.18 (0.08–0.28) [48]	9.29 (5.11–13.47) [52][a]	
		0.14 (0.11–0.16) [56]	11.84 (7.20–16.48) [53][a]	
			13.74 (10.90–16.58) [54][a]	
			0.74 (0.59–0.89) [56]	
			0.94 (0.56–1.32) [59][a]	
	AEs	0.65 (0.61–0.68) [32]	2.39 (2.23–2.55) [32]	0.21 (0.19–0.23) [32]
Women's football	Hours	0.35 (0.18–0.52) [42]	1.06 (0.33–1.79) [42]	0.22 (0.08–0.36) [42]
		0.62 (0.54–0.71) [44, 45]	3.28 (2.75–3.81) [45]	0.20 (0.15–0.26) [44]
		0.24 (0.13–0.35) [46]	1.21 (0.52–1.89) [50]	0.04 (0.00–0.09) [50]
		0.11 (0.02–0.20) [48]	11.67 (8.88–14.46) [51][a]	
		0.23 (0.14–0.46) [50]		
	AEs	0.67 (0.63–0.72) [36]	2.28 (2.11–2.44) [36]	0.20 (0.18–0.23) [36]
Men's rugby	Hours	0.58 (0.24–0.92) [91]	4.58 (3.61–5.56) [21][a]	0.12 (0.00–0.28) [91]
		1.60 (1.00–2.19) [92]	4.69 (1.63–7.75) [91]	0.05 (0.02–0.07) [95]
		0.52 (0.44–0.60) [95]	14.58 (9.18–19.99) [92]	
			6.55 (5.52–7.59) [95]	
	AEs	2.96 (2.41–3.50) [21][a]	6.10 (4.80–7.39) [21][a]	1.15 (0.73–1.58) [21][a]
		1.12 (0.93–1.31) [97][a]	9.64 (3.34–15.93) [99]	0.98 (0.00–2.08) [99]
		3.00 (1.30–4.69) [99]		
Women's rugby	Hours	1.36 (0.42–2.30) [98][a]	3.49 (2.56–4.41) [21][a]	
	AEs	1.99 (1.53–2.46) [21][a]	4.64 (3.41–5.86) [21][a]	0.67 (0.34–1.00) [21][a]
		0.60 (0.42–0.78) [97][a]		
Men's ice hockey	Hours	3.07 (2.15–3.99) [69][a]	11.67 (7.50–15.85) [69][a]	1.14 (0.52–1.75) [69][a]
			18.46 (15.82–21.11) [72][a]	
	AEs	0.71 (0.67–0.76) [61]	2.53 (2.35–2.72) [61]	0.20 (0.17–0.23) [61]
		0.65 (0.42–0.88) [70]		
		1.31 (0.78–1.85) [71]		
Women's ice hockey	Hours		3.39 (1.73–5.06) [73][a]	
	AEs	1.09 (0.87–1.31) [60]	3.24 (2.46–4.01) [60]	0.41 (0.26–0.57) [60]
		1.30 (0.53–2.06) [71]		
Men's American Football	AEs	1.18 (0.97–1.40) [86][a]	4.35 (1.34–7.36) [99]	0.82 (0.48–1.16) [99]
		1.04 (0.67–1.41) [99]		

HIID head injury incidence density, *CI* confidence interval, *AEs* athletic exposures

[a]Medical attention injury definition

[45] and 0.19 [42, 44], respectively. Only two studies reported HIID in football players per 1000 AEs with rates ranging from 0.20 [36] to 2.39 [32]. The average season, match and training incidence rates per 1000 AEs were 0.66, 2.34 and 0.21, respectively. In rugby HIID ranged from 0.05 [95] to 14.58 [92] per 1000 h and 0.60 [97] to 9.64 [99] per 1000 AEs. The median season, match and training incidence rates per 1000 h/AEs were 0.97/1.99 [21, 91, 98], 4.69/6.10 [21, 91] and 0.09/0.98 [21, 91, 99], respectively. In ice hockey, very little information on head injuries was available. Incidence rates per 1000 h/AEs ranged from 1.14/0.20 in training [61, 69] to 18.46/3.24 during match play [60, 72]. One study [69] reported a season incidence rate of 3.07 per 1000 h and the median

season HIID per 1000 AEs was 0.71 [61]. In American Football, there were no studies reporting incidence rates per 1000 h and two studies that assessed HIID per 1000 AEs; rates ranged from 0.82 in training [99] to 4.35 during match play [99].

Overall, training head injuries were rare when compared to match injuries in all sports; the median rate ratios in football, rugby and ice hockey were 15.81 [40, 44, 45], 9.84 [99] and 10.18 [69], respectively. In American Football only one study evaluated head injuries in both match and training; the rate ratio was 5.30 [99]. When comparing male and female players, no relevant differences of HIID were found in football [32, 36, 42, 44, 45]. In one study [21] male rugby players were more likely to incur a head injury with female/male rate ratios of 0.76 (match) and 0.58 (training). The opposite was observed in ice hockey according to studies by Agel et al. [60, 61], where female players had a 1.28 (match) and 2.05 (training) times higher HIID. However, studies by Tuominen et al. [72, 73] found male ice hockey players to have a notably higher match HIID with a female/male rate ratio of 0.18.

4 Discussion

4.1 Concussions

Summarising the evidence from both our qualitative and quantitative analysis on concussions, the highest incidence density rates were found consistently in rugby and the lowest in football. This was observed irrespective of exposure definition or context with a few exceptions; in our meta-analysis on training concussions per 1000 h, female football players had the highest incidence rates. However, the meta-analysis in men's rugby training only included two studies, one of which measured many hours of exposure and therefore carried considerable weight in the pooled incidence rate. Thus, the pooled CID for rugby training is mainly based on a single study [95] and should be interpreted with caution, even though this was a high-quality study. In line with these results, a review by Koh et al. [24] found the lowest incidence rates among the four sports in men's football, and the highest rates at the professional level in men's ice hockey and rugby. Of note, the results of this review were qualitative, as no meta-analysis was conducted, and the analysis included amateur players.

When comparing CID in men's and women's sport, our meta-analysis showed higher rates in female football and ice hockey players during match and training. Considering that international ice hockey rules do not allow body checking in women's ice hockey, while it is allowed in the men's game, this is surprising. In a review by Abrahams et al. [17], who also found higher CID rates in female football players, several reasons for an increased concussion susceptibility of female athletes are discussed; these include differences in neck strength and head mass [103], increased head-to-ball ratio [104] and increased willingness to report injuries [105]. The latter might be especially relevant for 'invisible' injuries such as concussions. When comparing match and training concussions, training concussions were rare across all sports, contexts and in both male and female players. Rate ratios should be interpreted carefully since very low training incidence rates led to great variation in rate ratios. The considerably higher concussion incidence in matches is likely due to higher intensity and body contact during matches, but may also reflect the absence of a trained professional to diagnose concussions during training [106].

4.2 Head Injuries

Head injury incidence varied dramatically across studies, which might be largely due to methodological differences. These include different injury definitions (TL, MA), and in addition some studies did not differentiate between head and neck injuries. This rendered an inter-study comparison very difficult and therefore no meta-analysis could be conducted on head injuries. However, the results of our qualitative analysis were similar to our findings on concussions, with the lowest HIID rates reported consistently in football, and the highest in rugby and ice hockey. Thus, while in rugby more players seem to incur or report a concussion, similar rates were observed when evaluating the incidence of all head injuries. This may be due to an actual difference in concussion incidence or reflect the success of the International Rugby Board's (World Rugby) continuing efforts to increase awareness and implement strict concussion policies regarding reporting and removal from play [106]. When comparing match and training, match head injuries were yet again much more frequent. Comparing male and female players, no consistent or meaningful differences could be found; however, there was little evidence available on which this conclusion is based.

4.3 Study Quality and Recommendations

There were several limitations regarding risk of bias in the reported incidence density rates. First, when looking more closely at the amount of exposure measured, a great number of studies were underpowered, especially in football, where concussions are relatively rare events. As an example, based on the pooled incidence rate in football matches on average one concussion is expected in roughly 2000 exposure hours. Assuming that at least ten events are needed to approximate the true lambda of any given population (and this is probably still a low number), studies

measuring less than 20,000 football match hours do not produce statistically meaningful CIDs. Second, often the sample population was not sufficiently described, particularly regarding number of participants and age. While the terms 'sample size' and 'reported athlete exposures' sometimes seem to be used interchangeably in epidemiological studies, it is not the same information and both should be reported. Third, the intervals between data collection were often too long. To reduce under-reporting and improve accuracy, injury data need to be collected every week for training exposure and every day during tournaments [101]. Fourth, varying injury definitions were used and concussion diagnosis often relied on the clinical expertise of the data collection personnel. Due to the fact that a concussion remains an injury that is often overlooked, a time-loss injury definition may lead to an underestimation of the true incidence. Future studies should use a concussion-specific injury definition beyond TL or MA and a comprehensive, standardised test battery for the diagnosis of concussion [28]. Finally, it would be desirable for injury data to be reported in a more consistent manner across studies, so future reviews will be able to base their conclusions on more data. As the two most common exposure definitions, hours and AEs, both have their advantages, ideally both should be evaluated and rates reported per 1000 exposures.

4.4 Limitations

The main limitation of this review is the methodological heterogeneity among included studies. Some results are based on very limited data, due to the aforementioned methodological differences, underpowered studies, lack of access to unpublished data and little research conducted in female players and in a training context. Moreover, we did not distinguish between different injury definitions in the meta-analysis; however, there were both TL and MA studies in all subgroups. Further, statistical heterogeneity was high in our meta-analysis. During the period encompassed by this review several rule changes and other concussion prevention strategies were implemented in all four sports. Thus, data from older studies may have inflated the reported CIDs. As head injuries other than concussions were not the main focus of this review, further reviews are needed to provide a more comprehensive analysis of HIID.

5 Conclusions

This systematic review provides a comprehensive estimate of CID in football, rugby, ice hockey and American Football; to the authors' knowledge this is the first meta-analysis comparing CID in contact sports. Several risk factors were taken into account, including sex and match or training context. Overall, the highest incidence rates were found in rugby and in a match context. In football and ice hockey, female players had an increased concussion risk compared to male players. More information is needed on concussion in women's contact sports, particularly regarding modifiable risk factors that could lead to an increased CID in female players. These factors may include differences in head-to-ball ratio in women's football, (neck) strength as well as rules/rules enforcement concerning heading, tackling and body checking. In order to produce statistically meaningful results, future studies of concussions in sport should focus on sufficient sample size and power. To avoid under-diagnosis particularly during training, professional monitoring, a concussion-specific injury definition and a comprehensive, standardised diagnostic procedure are recommended.

Compliance with Ethical Standards

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